

UDP

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1 udp	1
2 Directory Hierarchy	3
2.1 Directories	3
3 File Index	5
3.1 File List	5
4 Directory Documentation	7
4.1 build/CMakeFiles/4.3.2 Directory Reference	7
4.2 build Directory Reference	7
4.3 build/CMakeFiles Directory Reference	7
4.4 build/CMakeFiles/4.3.2/CompilerIdC Directory Reference	7
4.5 build/CMakeFiles/shared_udp.dir Directory Reference	7
4.6 build/CMakeFiles/static_udp.dir Directory Reference	8
5 File Documentation	9
5.1 build/CMakeFiles/4.3.2/CompilerIdC/CMakeCCompilerId.c File Reference	9
5.1.1 Macro Definition Documentation	10
5.1.1.1 __has_include	10
5.1.1.2 ARCHITECTURE_ID	10
5.1.1.3 C_STD_11	10
5.1.1.4 C_STD_17	10
5.1.1.5 C_STD_23	10
5.1.1.6 C_STD_99	10
5.1.1.7 C_VERSION	10
5.1.1.8 COMPILER_ID	10
5.1.1.9 DEC	11
5.1.1.10 HEX	11
5.1.1.11 PLATFORM_ID	11
5.1.1.12 STRINGIFY	11
5.1.1.13 STRINGIFY_HELPER	11
5.1.2 Function Documentation	11
5.1.2.1 main()	11
5.1.3 Variable Documentation	12
5.1.3.1 info_arch	12
5.1.3.2 info_compiler	12
5.1.3.3 info_language_extensions_default	12
5.1.3.4 info_language_standard_default	12
5.1.3.5 info_platform	12
5.2 build/CMakeFiles/shared_udp.dir/link.d File Reference	12
5.3 build/CMakeFiles/static_udp.dir/link.d File Reference	12
5.4 build/CMakeFiles/shared_udp.dir/udp.c.o.d File Reference	12
5.5 build/CMakeFiles/static_udp.dir/udp.c.o.d File Reference	12

5.6 README.md File Reference	12
5.7 udp.c File Reference	12
5.7.1 Function Documentation	13
5.7.1.1 main()	13
Index	15

Chapter 1

udp

Chapter 2

Directory Hierarchy

2.1 Directories

4.3.2	7
CompilerIdC	7
CMakeCCompilerId.c	9
build	7
CMakeFiles	7
4.3.2	7
CompilerIdC	7
CMakeCCompilerId.c	9
shared_udp.dir	7
link.d	12
udp.c.o.d	12
static_udp.dir	8
link.d	12
udp.c.o.d	12
CMakeFiles	7
4.3.2	7
CompilerIdC	7
CMakeCCompilerId.c	9
shared_udp.dir	7
link.d	12
udp.c.o.d	12
static_udp.dir	8
link.d	12
udp.c.o.d	12
CompilerIdC	7
CMakeCCompilerId.c	9
shared_udp.dir	7
link.d	12
udp.c.o.d	12
static_udp.dir	8
link.d	12
udp.c.o.d	12

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

udp.c	12
build/CMakeFiles/4.3.2/CompilerIdC/ CMakeCCompilerId.c	9
build/CMakeFiles/shared_udp.dir/ link.d	12
build/CMakeFiles/shared_udp.dir/ udp.c.o.d	12
build/CMakeFiles/static_udp.dir/ link.d	12
build/CMakeFiles/static_udp.dir/ udp.c.o.d	12

Chapter 4

Directory Documentation

4.1 build/CMakeFiles/4.3.2 Directory Reference

Directories

- directory [CompilerIdC](#)

4.2 build Directory Reference

Directories

- directory [CMakeFiles](#)

4.3 build/CMakeFiles Directory Reference

Directories

- directory [4.3.2](#)
- directory [shared_udp.dir](#)
- directory [static_udp.dir](#)

4.4 build/CMakeFiles/4.3.2/CompilerIdC Directory Reference

Files

- file [CMakeCCompilerId.c](#)

4.5 build/CMakeFiles/shared_udp.dir Directory Reference

Files

- file [link.d](#)
- file [udp.c.o.d](#)

4.6 build/CMakeFiles/static_udp.dir Directory Reference

Files

- file [link.d](#)
- file [udp.c.o.d](#)

Chapter 5

File Documentation

5.1 build/CMakeFiles/4.3.2/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- `#define __has_include(x)`
- `#define COMPILER_ID ""`
- `#define STRINGIFY_HELPER(X)`
- `#define STRINGIFY(X)`
- `#define PLATFORM_ID`
- `#define ARCHITECTURE_ID`
- `#define DEC(n)`
- `#define HEX(n)`
- `#define C_STD_99 199901L`
- `#define C_STD_11 201112L`
- `#define C_STD_17 201710L`
- `#define C_STD_23 202311L`
- `#define C_VERSION`

Functions

- `int main (int argc, char *argv[ ])`

Variables

- `char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"`
- `char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"`
- `char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"`
- `const char * info_language_standard_default`
- `const char * info_language_extensions_default`

5.1.1 Macro Definition Documentation

5.1.1.1 `__has_include`

```
#define __has_include(  
    x)
```

Value:

0

5.1.1.2 `ARCHITECTURE_ID`

```
#define ARCHITECTURE_ID
```

5.1.1.3 `C_STD_11`

```
#define C_STD_11 201112L
```

5.1.1.4 `C_STD_17`

```
#define C_STD_17 201710L
```

5.1.1.5 `C_STD_23`

```
#define C_STD_23 202311L
```

5.1.1.6 `C_STD_99`

```
#define C_STD_99 199901L
```

5.1.1.7 `C_VERSION`

```
#define C_VERSION
```

5.1.1.8 `COMPILER_ID`

```
#define COMPILER_ID ""
```

5.1.1.9 DEC

```
#define DEC(
    n)
```

Value:

```
('0' + ((n) / 10000000) % 10), \
('0' + ((n) / 1000000) % 10), \
('0' + ((n) / 100000) % 10), \
('0' + ((n) / 10000) % 10), \
('0' + ((n) / 1000) % 10), \
('0' + ((n) / 100) % 10), \
('0' + ((n) / 10) % 10), \
('0' + ((n) % 10))
```

5.1.1.10 HEX

```
#define HEX(
    n)
```

Value:

```
('0' + ((n) >> 28 & 0xF)), \
('0' + ((n) >> 24 & 0xF)), \
('0' + ((n) >> 20 & 0xF)), \
('0' + ((n) >> 16 & 0xF)), \
('0' + ((n) >> 12 & 0xF)), \
('0' + ((n) >> 8 & 0xF)), \
('0' + ((n) >> 4 & 0xF)), \
('0' + ((n) & 0xF))
```

5.1.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

5.1.1.12 STRINGIFY

```
#define STRINGIFY(
    X)
```

Value:

```
STRINGIFY_HELPER(X)
```

5.1.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
    X)
```

Value:

```
#X
```

5.1.2 Function Documentation

5.1.2.1 main()

```
int main (
    int argc,
    char * argv[])
```

5.1.3 Variable Documentation

5.1.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

5.1.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

5.1.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
= "INFO" ":" "extensions_default["  
  
    "OFF"  
"]"
```

5.1.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
"INFO" ":" "standard_default[" C_VERSION  
    =  
"]"
```

5.1.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

5.2 build/CMakeFiles/shared_udp.dir/link.d File Reference

5.3 build/CMakeFiles/static_udp.dir/link.d File Reference

5.4 build/CMakeFiles/shared_udp.dir/udp.c.o.d File Reference

5.5 build/CMakeFiles/static_udp.dir/udp.c.o.d File Reference

5.6 README.md File Reference

5.7 udp.c File Reference

```
#include <udp.h>  
#include <getopt.h>  
#include <stddef.h>  
#include <string.h>  
#include <stdbool.h>
```


Functions

- `int main (int argc, char **argv)`
Entry point for the UDP packet crafting and transmission utility.

5.7.1 Function Documentation

5.7.1.1 main()

```
int main (  
    int argc,  
    char ** argv)
```

Entry point for the UDP packet crafting and transmission utility.

This program parses command-line arguments to build a custom UDP packet. It allows configuration of MAC addresses, IP addresses, ports, and payload data.

Command-line Options:

- `-w, --stdio` Read payload data from standard input.
- `-e, --print` Print the packet structure to the console before sending.
- `-i, --ip-address-destination` Set the destination IPv4 address.
- `-s, --ip-address-source` Set the source IPv4 address.
- `-p, --port-destination` Set the destination UDP port.
- `-o, --port-source` Set the source UDP port.
- `-n, --interface` Specify the network interface (e.g., eth0).
- `-f, --file` Read payload data from a specified file.
- `-m, --mac-address-destination` Set the destination MAC address.
- `-a, --mac-address-source` Set the source MAC address.

Payload Logic:

1. If `-w` or `-f` is provided, the data is pulled from those sources.
2. If no source flag is provided, the program concatenates all remaining positional arguments (`argv`) into a single space-separated string payload.

Parameters

<code>argc</code>	The number of command-line arguments.
<code>argv</code>	The array of command-line argument strings.

Returns

- 0 on successful transmission.
- -1 if memory allocation fails (`init_udp_pack`).
- Non-zero error code if packet configuration or sending fails.

Index

- [__has_include](#)
[CMakeCCompilerId.c, 10](#)
- [ARCHITECTURE_ID](#)
[CMakeCCompilerId.c, 10](#)
- [build](#) [Directory Reference, 7](#)
- [build/CMakeFiles](#) [Directory Reference, 7](#)
- [build/CMakeFiles/4.3.2](#) [Directory Reference, 7](#)
- [build/CMakeFiles/4.3.2/CompilerIdC](#) [Directory Reference, 7](#)
- [build/CMakeFiles/4.3.2/CompilerIdC/CMakeCCompilerId.c](#), [9](#)
- [build/CMakeFiles/shared_udp.dir](#) [Directory Reference, 7](#)
- [build/CMakeFiles/shared_udp.dir/link.d](#), [12](#)
- [build/CMakeFiles/shared_udp.dir/udp.c.o.d](#), [12](#)
- [build/CMakeFiles/static_udp.dir](#) [Directory Reference, 8](#)
- [build/CMakeFiles/static_udp.dir/link.d](#), [12](#)
- [build/CMakeFiles/static_udp.dir/udp.c.o.d](#), [12](#)
- [C_STD_11](#)
[CMakeCCompilerId.c, 10](#)
- [C_STD_17](#)
[CMakeCCompilerId.c, 10](#)
- [C_STD_23](#)
[CMakeCCompilerId.c, 10](#)
- [C_STD_99](#)
[CMakeCCompilerId.c, 10](#)
- [C_VERSION](#)
[CMakeCCompilerId.c, 10](#)
- [CMakeCCompilerId.c](#)
 - [__has_include, 10](#)
 - [ARCHITECTURE_ID, 10](#)
 - [C_STD_11, 10](#)
 - [C_STD_17, 10](#)
 - [C_STD_23, 10](#)
 - [C_STD_99, 10](#)
 - [C_VERSION, 10](#)
 - [COMPILER_ID, 10](#)
 - [DEC, 10](#)
 - [HEX, 11](#)
 - [info_arch, 12](#)
 - [info_compiler, 12](#)
 - [info_language_extensions_default, 12](#)
 - [info_language_standard_default, 12](#)
 - [info_platform, 12](#)
 - [main, 11](#)
 - [PLATFORM_ID, 11](#)
 - [STRINGIFY, 11](#)
 - [STRINGIFY_HELPER, 11](#)
- [COMPILER_ID](#)
[CMakeCCompilerId.c, 10](#)
- [DEC](#)
[CMakeCCompilerId.c, 10](#)
- [HEX](#)
[CMakeCCompilerId.c, 11](#)
- [info_arch](#)
[CMakeCCompilerId.c, 12](#)
- [info_compiler](#)
[CMakeCCompilerId.c, 12](#)
- [info_language_extensions_default](#)
[CMakeCCompilerId.c, 12](#)
- [info_language_standard_default](#)
[CMakeCCompilerId.c, 12](#)
- [info_platform](#)
[CMakeCCompilerId.c, 12](#)
- [main](#)
[CMakeCCompilerId.c, 11](#)
[udp.c, 13](#)
- [PLATFORM_ID](#)
[CMakeCCompilerId.c, 11](#)
- [README.md, 12](#)
- [STRINGIFY](#)
[CMakeCCompilerId.c, 11](#)
- [STRINGIFY_HELPER](#)
[CMakeCCompilerId.c, 11](#)
- [udp, 1](#)
- [udp.c, 12](#)
 - [main, 13](#)